3(ai)	Gas is ideal as the graph is a straight line through the origin, satisfying the ideal gas equation $pV = NkT$	[1]
3(aii)	$pV = NkT$ $N = pV / kT$ = gradient of graph $= 270 / 8.0 \times 10^{-21}$ or substitute a data point $n = N/N_A$ $= (3.4 \times 10^{22})/ (6.02 \times 10^{23})$ $= 0.056$ mol	[1] [1]
3(b)	$\frac{1}{2} m\langle c^2 \rangle = 3/2\ kT$ When $pV = 270$ J, $kT = 8.0 \times 10^{-21}$ J $\frac{1}{2} m (1900^2) = 1.5 (8.0 \times 10^{-21})$ $m = 6.65 \times 10^{-27}$ kg $= (6.65 \times 10^{-27}) / (1.66 \times 10^{-27})$ $= 4.0\ u$	 [1] [1] [1]